

M13 Batch 4 — Full Set

M13.3 Autoflight (ATA 22) • PDF pages 232–321

Flight director and take-off mode, auto-throttle, yaw damping, all weather operation and the ILS categories, system redundancy, helicopter stability augmentation.

311 questions • Level 3: 104 • Level 2: 153 • Level 1: 54

● marks a question built on a fact your instructor highlighted or underlined in the manual (258 of 311). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Flight director, roll bar and pitch bar

PDF pages 232–238 • 31 questions

LEVEL 1 — RECALL

1. Which two tasks the flight director accomplishes ●
 - a. Commands for manual control and monitoring of autopilot activities
 - b. Commands for manual control and automatic trim of the pitch axis
 - c. Monitoring of autopilot activities and selection of the thrust rating
2. During which phases a yaw bar is displayed and the roll bar is removed ●
 - a. Flare below 50 ft radio height, climb and take-off
 - b. Align below 30 ft radio height, rollout and approach
 - c. Flare below 30 ft radio height, rollout and take-off
3. Which bars display the attitude reference commands in an ADI or PFD ●
 - a. A pitch bar, a track bar and eventually a roll bar
 - b. A roll bar, a speed bar and eventually a yaw bar
 - c. A pitch bar, a roll bar and eventually a yaw bar
4. Which three speeds the pilots determine at flight preparation for the take off phase ●
 - a. V1, V2 and VLS
 - b. V1, VR and V2
 - c. VR, V2 and VMAX
5. At which height above the runway the take off safety speed has to be obtained with one engine inoperative ●
 - a. 35 ft
 - b. 30 ft
 - c. 50 ft

LEVEL 2 — UNDERSTAND AND EXPLAIN

6. Why the flight director indicators are overlaid on the artificial horizon ●
 - a. They display attitude commands affecting the rotary degrees of freedom
 - b. They display track commands affecting the lateral degrees of freedom
 - c. They display speed commands affecting the linear degrees of freedom

7. An aircraft with a crossbar flight director is following its defined flight path. Where do the two bars lie?
 - a. They rest at the upper edge of the own aircraft symbol
 - b. They lie apart at an equal distance above the horizon line
 - c. They intersect at the middle mark of the own aircraft symbol
8. Which rotations the yaw bar commands ●
 - a. Rotations around the vertical axis only
 - b. Rotations around the lateral axis only
 - c. Rotations around the longitudinal axis only
9. The yaw bar in the ADI deflects during the take off run. Which control inputs does the pilot make? ●
 - a. Elevator or stabiliser trim inputs
 - b. Rudder or nose wheel steering inputs
 - c. Aileron or roll spoiler inputs
10. Below 50 ft radio altitude on an autoland the flare, align and rollout modes are selected automatically. What part does the flight director then play? ●
 - a. It is removed from the display until roll out has ended
 - b. It serves only for commanding manual control inputs
 - c. It serves only for monitoring the autopilot
11. The autopilot is engaged and a lateral mode is chosen on the control panel. What happens to the flight director? ●
 - a. The selected mode also holds for the flight director
 - b. The flight director bars are biased out of view
 - c. The flight director reverts to its take off mode
12. During a flight path mode the aircraft adopts the commanded attitude. What do the command bars do?
 - a. They return immediately to their centre position
 - b. They stay deflected until the mode is deselected
 - c. They move to a steady offset from the centre
13. The pitch bar deflects upward. Which input does it call for? ●
 - a. Pull elevator, pitch up
 - b. Right aileron, turn right
 - c. Push elevator, pitch down
14. Which input a deflection of the roll bar to the right signifies ●
 - a. Left aileron, turn left
 - b. Right rudder, yaw right
 - c. Right aileron, turn right
15. The yaw bar deflects to the left. Which input does that command? ●
 - a. Left aileron wheel, turn left
 - b. Left rudder pedal, yaw left
 - c. Right rudder pedal, yaw right
16. The roll bar in the ADI deflects. Which autopilot signal does that deflection correspond to? ●
 - a. The attitude error computed in the roll outer control loop
 - b. The signal fed to the roll actuator in the outer heading loop
 - c. The signal fed to the roll actuator in the inner roll loop

17. Which action removes a roll attitude error that the flight director indicates
 - a. A reset made in the outer heading loop
 - b. A control input made by the pilot himself
 - c. A servo input made by the roll actuator
18. Why must an aircraft be able to hold different steady pitch attitudes?
 - a. They depend on thrust rating, bleed air load and temperature
 - b. They depend on runway heading, visibility and flap setting
 - c. They depend on airspeed, centre of gravity and gross weight
19. Why distances and airspeeds are laid down in the airworthy standards ●
 - a. To keep the engine ratings within their derated temperature limits
 - b. To guarantee a sufficient safety margin in all flight phases
 - c. To allow a take off at the lowest possible runway visibility
20. Why an aircraft with a tricycle gear has to be rotated at VR ●
 - a. It accelerates at a low angle of attack and needs more lift
 - b. It accelerates on its nose gear and has to unload that gear
 - c. It accelerates at a high angle of attack and needs less drag

LEVEL 3 — APPLY AND REASON

21. A flight director shows a roll attitude error and the pilot moves the wheel to follow it. Which part of the loop structure has the pilot just closed?
 - a. The inner control loop
 - b. The attitude limiter path
 - c. The outer heading loop
22. Why integration has to be avoided when a vertical mode drives the flight director pitch bar
 - a. The behaviour of the pilot cannot be predicted
 - b. The pitch bar signal is inverted before it is displayed
 - c. The outer loop error is already proportional to attitude
23. What a delayed pilot reaction would do to the integrator in the outer loop
 - a. Drive it towards saturation and wind up the pitch bar
 - b. Hold it at a constant value and freeze the pitch bar
 - c. Drive it towards zero and centre the pitch bar
24. Why the outer loop error cannot be used directly as the pitch attitude reference for the inner loop
 - a. A zero error would then demand a pitch attitude of 20 degrees
 - b. A zero error would then demand a constant pitch rate
 - c. A zero error would then demand a pitch attitude of 0 degrees
25. The outer loop error is zero and the aircraft is held at a pitch attitude inside the suppression range. Where does the flight director pitch bar sit? ●
 - a. Deflected fully up
 - b. Deflected fully down
 - c. In its neutral position
26. The aircraft pitches up beyond the suppression range. What is added to the pitch bar deflection? ●
 - a. An opposing command proportional to the exceeding angle
 - b. A command proportional only to the outer loop error
 - c. A fixed command until the attitude re-enters the range

27. The pilot makes no input in answer to a short-term pitch bar deflection. What happens to the bar?
 - a. It returns to its initial position again
 - b. It stays deflected until the mode is reselected
 - c. It increases its deflection steadily
28. The pilot follows a short-term pitch bar deflection caused by a gust inside the suppression range. What keeps that deflection from winding up?
 - a. The integrator in the outer control loop
 - b. The washout in the pitch rate feedback path
 - c. The attitude limiter at the pitch bar output
29. An engine fails while the aircraft is accelerating and is still below V1. What must be done? ●
 - a. The aircraft has to be rotated at once
 - b. The take off has to be continued
 - c. The take off has to be discontinued
30. A caution is generated by the warning system just after V1 has been passed. What must be done? ●
 - a. The take off has to be discontinued
 - b. The aircraft has to be held on its main gear
 - c. The take off has to be continued
31. An aircraft is in a V2 climb after an engine failure and below the acceleration altitude. What is the state of the gear and the flaps? ●
 - a. Gear retracted, take off flaps already retracted
 - b. Gear extended, take off flaps already retracted
 - c. Gear retracted, take off flaps still extended

Answers — Flight director, roll bar and pitch bar

1. a The page names exactly these two tasks: commands for manual control, and monitoring of the autopilot. (L1, PDF p. 232)
2. c Those are the three phases printed for yaw bar display. (L1, PDF p. 232)
3. c Pitch and roll bars, and possibly a yaw bar, carry the attitude reference commands. (L1, PDF p. 233)
4. b The three take off speeds named are V1, VR and V2. (L1, PDF p. 238)
5. a V2 is defined at 35 ft above the runway surface. (L1, PDF p. 238)
6. a Because the commands concern the rotary degrees of freedom they belong on the artificial horizon. (L2, PDF p. 232)
7. c On path the roll bar crosses the pitch bar at the middle mark of the aircraft symbol. (L2, PDF p. 232)
8. a The yaw bar is restricted to rotations about the vertical axis. (L2, PDF p. 232)
9. b Yaw bar commands are flown with rudder or nose wheel steering. (L2, PDF p. 232)
10. c In the autoland modes the flight director is a monitoring display only. (L2, PDF p. 232)
11. a A mode selected with the autopilot engaged applies to the flight director as well. (L2, PDF p. 232)
12. a The bars show attitude error only, so they centre as soon as the error is gone. (L2, PDF p. 233)
13. a An upward pitch bar asks for elevator back pressure to pitch up. (L2, PDF p. 233)
14. c A roll bar to the right asks for right aileron and a turn to the right. (L2, PDF p. 233)
15. b A yaw bar to the left asks for left rudder pedal to yaw left. (L2, PDF p. 233)
16. c The bar is driven by the same signal the autopilot would send to its roll actuator in the inner loop. (L2, PDF p. 234)
17. b The flight director only displays the command; the correction comes from the pilot's own input. (L2, PDF p. 234)
18. c Airspeed, CG position and gross weight each change the steady attitude required. (L2, PDF p. 236)
19. b They exist to give a sufficient safety margin in every flight phase. (L2, PDF p. 238)
20. a The ground run is at a low angle of attack, so rotation is needed to make enough lift. (L2, PDF p. 238)

- 21. a In flight director operation the pilot himself closes the inner loop that the autopilot servo would otherwise close. (L3, PDF p. 234)
- 22. a A pilot may react late or not at all, so an integrator cannot be used. (L3, PDF p. 236)
- 23. a The integrator saturates, the bar winds up and the command becomes useless. (L3, PDF p. 236)
- 24. c With the outer loop error as reference, zero error would force the attitude to zero as well. (L3, PDF p. 236)
- 25. c Eight degrees up lies inside the printed suppression range, so with no outer loop error the bar rests neutral. (L3, PDF p. 236)
- 26. a Outside the range an opposing command sized by the exceeding angle is added to the bar. (L3, PDF p. 236)
- 27. a The washout lets the deflection decay back to the initial position. (L3, PDF p. 236)
- 28. b The washout makes the pitch rate feedback decay, so no lasting build-up occurs. (L3, PDF p. 236)
- 29. c Before V1 an engine failure obliges the crew to stop the take off. (L3, PDF p. 238)
- 30. c At or after V1 the take off is continued whatever the warning. (L3, PDF p. 238)
- 31. c The gear comes up but the take off flaps stay out until the acceleration altitude. (L3, PDF p. 238)

Take-off mode

PDF pages 240–246 · 29 questions

LEVEL 1 — RECALL

32. Which system the take off mode is exclusively reserved for ●
 - a. The flight director
 - b. The auto-throttle
 - c. The autopilot
33. The take off mode has just been selected on the ground. Which flight director channels are activated? ●
 - a. The pitch channel and the roll channel
 - b. The pitch channel and the yaw channel
 - c. The roll channel and the yaw channel
34. At which typical airspeed the rotation attitude is indicated on the pitch bar during the acceleration ●
 - a. Typically 80 kts
 - b. Typically 60 kts
 - c. Typically 100 kts
35. From which parameters the potential flight path angle is calculated ●
 - a. Longitudinal acceleration, pitch rate and radio altitude
 - b. Longitudinal and vertical acceleration and angle of attack
 - c. Vertical acceleration, airspeed and angle of attack
36. How the actual flight path angle is obtained ●
 - a. From the difference between pitch attitude and angle of attack
 - b. From the difference between pitch attitude and the threshold angle
 - c. From the sum of the pitch attitude and the angle of attack

LEVEL 2 — UNDERSTAND AND EXPLAIN

37. In which mode an armed auto-throttle system stays throughout the take off phase ●
 - a. Speed mode
 - b. Thrust mode
 - c. Mach mode
38. A take off is to be made at a visibility between 125 m and 75 m. Which indication gives the pilot his lateral guidance? ●
 - a. A roll bar controlled by the runway heading signal
 - b. A pitch bar controlled by the radio altitude signal
 - c. A yaw bar controlled by the localizer deviation signal
39. Why supplemental display systems are fitted for low visibility take offs ●
 - a. They replace the artificial horizon during the take off run
 - b. They avoid a change between head-up and head-down views
 - c. They allow the roll bar to be shown below 30 ft radio height
40. How the flight director is switched into take off mode ●
 - a. Pressing a NAV switch or pushing the levers into the idle detent
 - b. Arming the auto-thrust and selecting the departure runway by hand
 - c. Pressing a TOGA switch or pushing the levers into the TOGA detent

41. The aircraft passes 30 ft radio height with no FMS flight plan active. Which lateral guidance does the roll bar give? ●
- It maintains the runway centre line
 - It maintains the runway track
 - It maintains the heading entered on the control panel
42. When the pitch attitude to be adopted at VR is shown on the pitch bar ●
- As soon as the take off mode is selected, or at a certain airspeed
 - As soon as the gear struts extend, or at 35 ft above the runway
 - As soon as the thrust levers reach the TOGA detent, or at lift off
43. Which airspeed the flight director commands for the initial climb at a normal take off ●
- An actual airspeed 10 to 20 kts above V2
 - An actual airspeed 10 to 20 kts above VR
 - An actual airspeed equal to V2
44. A take off run is in progress. Which reference does the pilot use to find the point of rotation?
- Pitch bar position only
 - Airspeed only
 - Radio height only
45. Which two inputs the flight director needs for yaw bar control up to 30 ft radio height ●
- Runway heading and localizer deviation
 - Runway heading and radio altitude
 - Localizer deviation and ground speed
46. Why pitch attitude replaces angle of attack in that calculation during the take off phase ●
- Because of the air flow conditions at that time
 - Because the angle of attack vanes are not yet heated
 - Because the inertial system is not yet fully aligned
47. Which autopilot mode the flight director take off yaw control mode is equivalent to ●
- The autoland align mode
 - The autoland flare mode
 - The autoland roll-out mode
48. Which input tells the yaw controller the direction in which the aircraft should be pointing? ●
- The supplied runway heading
 - The localizer deviation signal
 - The radio height signal
49. Which input lets the yaw controller recognise a departure from the runway centre line? ●
- The supplied runway heading
 - The localizer deviation signal
 - The nose wheel steering position

LEVEL 3 — APPLY AND REASON

50. The aircraft climbs through a radio altitude of 30 ft after lift off. What changes in the flight director? ●
- The yaw channel is deactivated and the roll channel activated
 - The pitch channel is deactivated and the roll channel activated
 - The roll channel is deactivated and the yaw channel activated

51. Which guidance the yaw bar gives from brake release up to a radio height of 30 ft ●
- A return to the runway track after a lateral mode change
 - A return to the runway centre line at any deviation
 - A turn onto the standard instrument departure
52. An FMS flight plan is active as the aircraft climbs through 30 ft radio height. What does the lateral command follow? ●
- The standard instrument departure
 - The localizer of the departure runway
 - The runway centre line until 1500 ft
53. An engine fails after V1. Which pitch attitude does the flight director command? ●
- One that holds the rotation attitude until the acceleration altitude
 - One that reaches the safety speed at 35 ft and then holds it
 - One that reaches the safety speed plus 15 kts and then holds it
54. The pitch bar already shows a rotation attitude while the aircraft is still accelerating below VR. What must the pilot do?
- Rotate the aircraft into the pitch bar
 - Reduce thrust until the rotation speed is reached
 - Disregard that pitch bar command
55. The take off runway has been selected at flight preparation. How does the yaw channel obtain its two inputs?
- From the IRS, which supplies the track and the drift angle
 - From the ADC, which supplies the heading and the deviation
 - From the FMS, which tunes the ILS and supplies the heading
56. Heading mode is selected for lateral guidance after the yaw bar has been removed. Which heading is followed? ●
- The heading at its activation or one entered on the panel
 - The heading of the standard instrument departure leg
 - The runway heading supplied by the navigation database
57. During take off the potential flight path angle drops below the threshold while the speed is already above V2. What does the autoflight system do? ●
- It raises the speed target by a further 15 kts
 - It reduces the speed target to the safety speed
 - It tries to maintain the speed held at engine failure
58. The potential flight path angle falls below the threshold while the aircraft is still below the safety speed. What becomes the speed target? ●
- VLS
 - VR
 - V2
59. Which difference remains between that flight director mode and the equivalent autopilot mode ●
- The rudder is driven directly by the yaw servo
 - The localizer deviation signal is not made use of
 - The pilot is actively involved in the control loop

60. The pilot flies the yaw bar commands accurately during the take off run. What happens to the localizer deviation and the heading error? ●
- Both are replaced by a roll demand
 - Both of them are progressively minimized
 - Both are frozen at their present values

Answers — Take-off mode

32. a The take off mode belongs to the flight director alone. (L1, PDF p. 240)
33. b Selection of the take off mode brings up the pitch and yaw channels. (L1, PDF p. 240)
34. b The printed typical value is 60 kts. (L1, PDF p. 242)
35. b Those three parameters are the printed inputs to the potential flight path angle. (L1, PDF p. 244)
36. a It is the pitch attitude angle less the angle of attack. (L1, PDF p. 244)
37. b Through the whole take off the armed system remains in thrust mode. (L2, PDF p. 240)
38. c In that visibility band the yaw bar driven by localizer deviation provides the lateral guidance. (L2, PDF p. 240)
39. b They let the pilot see the guidance while still looking at the runway lighting. (L2, PDF p. 240)
40. c Either the TOGA switch or the TOGA detent selects the mode. (L2, PDF p. 242)
41. b Above 30 ft the roll bar holds the runway track until another lateral mode is chosen. (L2, PDF p. 242)
42. a Two implementations exist: at mode selection, or on passing a certain airspeed. (L2, PDF p. 242)
43. a On all engines the initial climb is flown 10 to 20 kts above V2. (L2, PDF p. 242)
44. b The rotation point is set by airspeed alone, read from the ASI. (L2, PDF p. 242)
45. a Yaw bar control needs the runway heading and the localizer deviation. (L2, PDF p. 242)
46. a The air flow conditions at take off make the angle of attack unusable, so pitch attitude is taken. (L2, PDF p. 244)
47. c It is the same control arrangement as the autoland roll-out mode. (L2, PDF p. 246)
48. a Alignment with the runway comes from the supplied runway heading. (L2, PDF p. 246)
49. b Deviations from the centre line are seen in the localizer deviation signal. (L2, PDF p. 246)
50. a At 30 ft the yaw channel drops out and the roll channel comes in at the same moment. (L3, PDF p. 240)
51. b Up to 30 ft the yaw bar works the aircraft back onto the runway centre line. (L3, PDF p. 242)
52. a With an active flight plan the lateral command follows the SID instead of the runway track. (L3, PDF p. 242)
53. b After an engine failure the commanded attitude reaches V2 at 35 ft and keeps it through the first and second segment climb. (L3, PDF p. 242)
54. c The flight director cannot find the rotation point, so a command before VR is ignored. (L3, PDF p. 242)
55. c The FMS tunes the ILS for the localizer deviation and reads the runway heading from its database. (L3, PDF p. 242)
56. a Heading mode follows either the heading at activation or the heading set on the control panel. (L3, PDF p. 242)
57. c Above V2 the system holds the speed the aircraft had at the failure instead of reducing the target. (L3, PDF p. 244)
58. c The target is automatically reduced to V2. (L3, PDF p. 244)
59. c The one difference is that the pilot sits in the loop. (L3, PDF p. 246)
60. b Following the bar drives both the lateral deviation and the heading error towards zero. (L3, PDF p. 246)

Auto-throttle: thrust, speed and Mach modes

PDF pages 248–262 · 60 questions

LEVEL 1 — RECALL

61. Which computer function determines the thrust ratings ●
 - a. The power management function of the engine computers
 - b. The thrust limit function of the flight management computers
 - c. The envelope protection function of the flight control computers
62. In which two areas an auto-throttle or auto-thrust system can relieve the pilot ●
 - a. Thrust control or airspeed control
 - b. Airspeed control or lateral guidance
 - c. Thrust control or pitch trim control
63. Which two modes of operation an auto-throttle or auto-thrust system provides ●
 - a. Speed or Mach mode and retard mode
 - b. Thrust mode and alpha floor mode
 - c. Thrust mode and speed or Mach mode
64. For how long the maximum take off thrust rating is basically available ●
 - a. 15 minutes
 - b. 10 minutes
 - c. 5 minutes
65. Which rating is the highest thrust available without a time limit ●
 - a. Maximum take off thrust
 - b. Maximum continuous thrust
 - c. Maximum climb thrust
66. The auto-throttle system works in thrust mode provided the autopilot vertical mode is of which kind ●
 - a. An altitude hold mode
 - b. A speed or Mach mode
 - c. A vertical speed mode
67. The rearward throttle movement to the idle position during cruise-descent and flare carries which name ●
 - a. Rate feedback
 - b. Torque limiter
 - c. Retard bias
68. In an auto-thrust system the required rating mode is brought in by which action ●
 - a. A servo motor driving the thrust levers forward
 - b. The pilot setting the thrust levers by hand
 - c. The flight management system commanding the levers
69. In speed or Mach mode the auto-throttle system holds which quantity ●
 - a. A pre-selected rating for the flight phase
 - b. A pre-selected airspeed or Mach number
 - c. A pre-selected Mach trim setting

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 70.** What the auto-thrust system does in place of moving the throttles ●
- It controls the throttle cables over the FADEC
 - It controls the thrust over the servo clutch
 - It controls the thrust over a data bus
- 71.** An aircraft has an auto-thrust system fitted. How can its throttles be moved? ●
- Manually or by the servomotor
 - Only automatically
 - Only manually
- 72.** What the position of the throttles determines on an aircraft with an auto-thrust system ●
- The reference speed used by the speed mode of the system
 - The thrust rating and so the limits for that flight phase
 - The thrust demand sent straight to the fuel control unit
- 73.** What supported the development of the auto-thrust system ●
- The introduction of the para visual display systems
 - The electronic control of modern turbofan engines
 - The derating of the turbine inlet temperature limits
- 74.** Which demands the auto-thrust system was introduced to meet
- Higher thrust ratings and a shorter take off run
 - Lower fuel burn and a wider speed envelope
 - Lower complexity and a better man/machine interface
- 75.** Which settings can be made in thrust mode ●
- Thrust ratings or alpha floor
 - Reference speed or Mach number
 - Thrust ratings or thrust idle
- 76.** What is obtained when thrust idle is set ●
- A thrust that just balances the aerodynamic drag
 - Only residual thrust from an idling engine
 - The lowest rating allowed for that flight phase
- 77.** The system is working in speed or Mach mode. Where does its reference speed come from? ●
- The autoflight control panel or the EEC
 - The air data computer or the FMS
 - The autoflight control panel or the FMS
- 78.** Which low speed value the protective functions keep the aircraft away from ●
- VR
 - V1
 - VLS
- 79.** With respect to what the overspeed protection works ●
- Altitude and the slat/flap or landing gear configuration
 - Angle of attack and the pitch attitude during the approach
 - Airspeed and the thrust rating set for that flight phase

- 80.** Which rule governs the control of airspeed by an autoflight system ●
- The system with the smaller error controls it at any moment
 - Only one system can control it exclusively at any moment
 - Both systems control it together at any moment
- 81.** Which system holds priority for airspeed control ●
- The auto-throttle or auto-thrust system
 - The autopilot or flight director pitch channel
 - The flight management system computer
- 82.** Why the maximum power available from a turbofan engine is reduced in service
- To keep the thrust within the capacity of the data bus
 - To allow the thrust to be measured directly in flight
 - To reach acceptable life cycles for highly stressed parts
- 83.** Which parameter limits the thermodynamic power of a turbofan engine ●
- The maximum low pressure rotor speed
 - The maximum engine pressure ratio
 - The maximum turbine inlet temperature
- 84.** The auto-throttle is working in thrust mode. Why are N1 or EPR used as the reference values? ●
- Thrust is a force and varies with the bleed air load in flight
 - Thrust is a force and is limited by the turbine inlet temperature
 - Thrust is a force and cannot be measured directly in flight
- 85.** Which period most authorities apart from the FAA accept for engine failure performance calculations
- 10 minutes of maximum take off thrust
 - 5 minutes of maximum take off thrust
 - 15 minutes of maximum continuous thrust
- 86.** Conditions allow a reduced thrust take off. Why is that reduction recommended? ●
- It shortens the take off run and lowers the noise footprint
 - It prolongs engine life and lowers the failure probability
 - It raises the thrust rating left available for the climb phase
- 87.** How go around thrust is defined ●
- The max climb rating corrected for the lower air density
 - The max continuous rating corrected for the higher altitude
 - The max take off rating corrected for the higher airspeed
- 88.** Idle thrust has to be set with care on every engine type, so what must it still guarantee? ●
- Spool-up to go-around power in the regulated time
 - Minimum fuel flow throughout the whole descent
 - Bleed air delivery at the regulated cabin pressure
- 89.** Idle thrust depends among other things on which characteristic of the engine fitted ●
- Its compression characteristic
 - Its acceleration characteristic
 - Its bleed extraction characteristic

90. The permanent rating calculation takes account of which two inputs
- Air data and bleed air demand
 - Thrust lever angle and fuel flow
 - Radio height and flap position
91. The crew has made no manual selection of the phase used for the rating — what else can supply it? ●
- The autothrottle computer unit
 - The engine control computer
 - The flight management system
92. Level change is engaged and the aircraft is descending — what thrust does the auto-throttle set? ●
- The climb rating
 - The take-off rating
 - Idle
93. Thrust mode is engaged on the auto-throttle — what type of control does the system itself apply? ●
- Open loop control
 - Closed loop control
 - Cascaded loop control
94. Thrust mode has just been selected — what does the determined thrust rating then serve as? ●
- The reference for fuel flow control by the hydromechanical unit
 - The reference for N2 closed loop control by the autothrottle computer
 - The reference for N1 closed loop control by the engine computer
95. The power management function works out the rating for which quantity ●
- Each engine bleed demand
 - Each defined flight phase
 - Each throttle position
96. In an auto-thrust system which device turns lever position into the signal sent to the engine control unit ●
- A rate gyro at the throttle quadrant
 - A torque limiter in the servo drive
 - An angle transducer at the levers
97. A technician finds no throttle servo motor on an auto-thrust aircraft — is that a defect? ●
- Yes, without it the retard bias cannot operate
 - Yes, the servo must drive the levers to the rating
 - No, the levers here are not driven automatically
98. The speed and Mach control structure is built on how many control loops
- A single loop
 - Two cascaded loops
 - Three parallel loops
99. The time response of the speed and Mach controller is fixed by which two factors ●
- Thrust change dynamics and aircraft aerodynamics
 - Air data computer lag and thrust lever angle range
 - Servo rate feedback and clutch torque limiter setting

100. The engine control input lever position is fed back to the servo — which effect is this meant to cut down? ●
- Hysteresis from the mechanical components
 - Overshoot from the longitudinal acceleration
 - Drift from the air data computer

LEVEL 3 — APPLY AND REASON

101. Which components of the earlier auto-throttle installation become superfluous with auto-thrust ●
- Servomotors, clutch mechanisms, cables and TOGA switches
 - Throttle levers, thrust limit displays and mode panels
 - Thrust rating computers, data buses and engine computers
102. Why does that relief only materialise when the pitch axis is also under autopilot control? ●
- Thrust changes trim the aircraft out and automatic trim is needed
 - The autopilot has to hold the attitude while the levers are moving
 - The pitch channel supplies the reference speed to the thrust computer
103. During an approach the angle of attack becomes extreme. Which protection acts and with which thrust? ●
- Overspeed protection, with idle thrust
 - Alpha floor protection, with go around thrust
 - Low speed protection, with climb thrust
104. Which limits are still observed while the system works in the speed or Mach mode ●
- The thrust ratings
 - The low speed limit
 - The take off rating
105. The pitch channel is given the task of holding a certain altitude. Which system then controls the airspeed? ●
- The flight management system computer
 - The pitch channel, which keeps its priority
 - The auto-throttle or auto-thrust system
106. The pitch channel is the system controlling the airspeed. In which mode is the auto-throttle then working? ●
- Speed mode, setting the reference airspeed
 - Thrust mode, setting a rating or idle thrust
 - Mach mode, setting the cruise Mach number
107. An aircraft takes off at a low mass from a long runway at a low air temperature. Which rating is recommended? ●
- Maximum continuous thrust
 - Flexible take off thrust
 - Maximum take off thrust
108. A go around is flown at go around thrust. For how long may that thrust be held? ●
- Until climb thrust is set at the acceleration altitude
 - For as long as the approach has to be repeated
 - For the same period as maximum take off thrust

- 109.** The maximum take off thrust period has elapsed but maximum thrust is still needed. Which rating is used?
- Maximum climb thrust
 - Maximum continuous thrust
 - Flexible take off thrust
- 110.** The servo drive is running the throttles forward in thrust mode — when does the movement stop?
- The power management value equals the rating reference
 - The thrust lever angle transducer reaches its forward stop
 - The engine reaches the commanded N1 at its control unit
- 111.** The aircraft climbs in thrust mode while pressure and temperature change — why do the throttles stay put?
- Both functions work from the same input data
 - The retard bias holds the levers against the idle stop
 - The servo drive is declutched above the transition altitude
- 112.** Two functions both work out ratings in thrust mode — what separates one from the other? ●
- One uses the flight phase, the other the lever position
 - One works only in flight, the other works only on the ground
 - One uses the air data, the other uses the bleed demand
- 113.** An aircraft has an auto-thrust system rather than an auto-throttle system — what is the practical difference? ●
- Its ratings are worked out without an air data system
 - Its thrust is set without an engine control system
 - Its thrust levers are not moved by a servo system
- 114.** An integrating element is used in this mode in order to obtain which result ●
- Less hysteresis in the linkage
 - A speed error of zero at the end
 - Faster throttle travel at large errors
- 115.** Hysteresis is not limited and the speed error grows — what happens to the rate of throttle movement?
- It rises in proportion to the error
 - It falls as the limiter takes over
 - It stays fixed until the error has gone
- 116.** Adding the engine control input position feedback to the servo makes which further item necessary ●
- A torque limiter behind the servo clutch
 - An electronic integrator ahead of the servo
 - A rate gyro ahead of the band pass filter unit
- 117.** The compromise between precise speed control and smooth thrust is reached by feeding back which signal
- Longitudinal acceleration
 - Actual pitch attitude angle
 - Actual throttle lever position

- 118.** A higher reference has been commanded and the aircraft is still accelerating with the error already gone — what stops an overshoot?
- The retard bias driving the throttles to idle
 - The acceleration bias on the speed error
 - The limiter holding the error at ten knots
- 119.** A head-on gust raises the indicated speed while the aircraft itself is slowing — what keeps the throttles still?
- The integrator inside the speed controller
 - The hysteresis of the throttle control cables
 - The acceleration bias on the airspeed error
- 120.** A gust suddenly raises the airspeed and no acceleration feedback is fitted — what would the throttles do?
- An advance of the throttles
 - No movement of the throttles
 - A cut back of the throttles

Answers — Auto-throttle: thrust, speed and Mach modes

- 61. a** Ratings come from the power management function in the EEC or ECU. (L1, PDF p. 248)
- 62. a** The two areas of relief are thrust control and airspeed control. (L1, PDF p. 248)
- 63. c** The system offers a thrust mode and a speed or Mach mode. (L1, PDF p. 250)
- 64. c** The printed basic limit is 5 minutes. (L1, PDF p. 254)
- 65. b** MCT is the highest thrust with no time limit attached. (L1, PDF p. 254)
- 66. b** Thrust mode is entered only while the pitch channel is holding speed or Mach. (L1, PDF p. 256)
- 67. c** The page names the drive back to idle in those two phases the retard bias. (L1, PDF p. 256)
- 68. b** The pilot picks the rating mode by hand on the levers. (L1, PDF p. 258)
- 69. b** The mode holds the speed or Mach value selected by the pilot or by the FMS. (L1, PDF p. 260)
- 70. c** Thrust is commanded over a data bus rather than by moving the levers. (L2, PDF p. 248)
- 71. c** With auto-thrust the levers move by hand only. (L2, PDF p. 248)
- 72. b** Lever position picks the rating, which sets the thrust limits for the phase. (L2, PDF p. 248)
- 73. b** FADEC engine control made the auto-thrust system possible, and the functions are often built into it. (L2, PDF p. 248)
- 74. c** It answered the demand for less complexity and a better man/machine interface. (L2, PDF p. 248)
- 75. c** Thrust mode sets either a rating, that is a thrust limit, or thrust idle. (L2, PDF p. 250)
- 76. b** Thrust idle leaves only the residual thrust of the idling engine. (L2, PDF p. 250)
- 77. c** The reference is either set on the control panel or supplied by the FMS. (L2, PDF p. 250)
- 78. c** VLS is the low speed limit the system protects. (L2, PDF p. 250)
- 79. a** Overspeed protection takes account of altitude and the slat/flap or gear configuration. (L2, PDF p. 250)
- 80. b** Airspeed is controlled by one system at a time, never two. (L2, PDF p. 252)
- 81. b** Priority lies with the pitch channel of the autopilot or flight director. (L2, PDF p. 252)
- 82. c** Continuous maximum thermodynamic power would wear the engine out quickly. (L2, PDF p. 252)
- 83. c** Turbine inlet temperature sets the thermodynamic ceiling, and is then derated. (L2, PDF p. 252)
- 84. c** Thrust itself cannot be measured in flight, so derived parameters stand in for it. (L2, PDF p. 252)
- 85. a** Those authorities allow 10 minutes of max take off thrust for that calculation. (L2, PDF p. 254)
- 86. b** Reduced thrust extends engine life and cuts the chance of a failure. (L2, PDF p. 254)
- 87. c** It is the max take off rating adjusted for the greater airspeed in a go around. (L2, PDF p. 254)
- 88. a** Idle may not be so low that the engine cannot spool up to go-around thrust in the required time. (L2, PDF p. 256)

- 89. b How fast the engine accelerates decides how low idle may be set. (L2, PDF p. 256)
- 90. a Those are the two inputs named for the rating calculation. (L2, PDF p. 256)
- 91. c The flight phase comes either from the pilot or from the FMS. (L2, PDF p. 256)
- 92. c In a level change descent the auto-throttle sets idle thrust. (L2, PDF p. 256)
- 93. a In thrust mode the auto-throttle itself works open loop. (L2, PDF p. 256)
- 94. c The rating becomes the N1 reference that the engine computer then closes the loop on. (L2, PDF p. 256)
- 95. c Power management maps throttle position to a rating, phase by phase is the other function. (L2, PDF p. 256)
- 96. c An angle transducer turns lever position into the TLA signal sent to the ECU. (L2, PDF p. 258)
- 97. c An auto-thrust system has no lever servo at all, so none is missing. (L2, PDF p. 258)
- 98. a Speed and Mach control uses one single loop. (L2, PDF p. 260)
- 99. a Those are the two things named as setting the controller response. (L2, PDF p. 260)
- 100. a The position feedback is there to reduce mechanical hysteresis. (L2, PDF p. 260)
- 101. a Those mechanical items and the TOGA switches drop out with auto-thrust. (L3, PDF p. 248)
- 102. a Every thrust change puts the aircraft out of trim, and the automatic trim comes from the autopilot pitch channel. (L3, PDF p. 248)
- 103. b Alpha floor protection answers an extreme angle of attack by applying go around thrust. (L3, PDF p. 250)
- 104. a Thrust ratings are observed even when speed is being controlled. (L3, PDF p. 250)
- 105. c The pitch channel delegates speed control once it is busy with another task. (L3, PDF p. 252)
- 106. b When the pitch channel has the speed, the thrust system sits in thrust mode on a rating or idle. (L3, PDF p. 252)
- 107. b Those conditions are exactly the case for the reduced, flexible take off rating. (L3, PDF p. 254)
- 108. c Go around thrust carries the same time limit as maximum take off thrust. (L3, PDF p. 254)
- 109. b That abnormal case is one of the two uses printed for maximum continuous thrust. (L3, PDF p. 254)
- 110. a The servo runs until the fed back power management value cancels the rating reference in the summing point. (L3, PDF p. 256)
- 111. a Both values move together, so their difference stays zero and no servo signal is produced. (L3, PDF p. 256)
- 112. a Rating calculation is keyed to the flight phase and power management to lever position. (L3, PDF p. 256)
- 113. c The lever servo is exactly what an auto-thrust system does without. (L3, PDF p. 258)
- 114. b Only an I-controller drives the remaining error to zero. (L3, PDF p. 260)
- 115. a Rate feedback makes throttle speed directly proportional to the error. (L3, PDF p. 260)
- 116. b The feedback destroys the servo's own integrating action, so an electronic integrator must replace it. (L3, PDF p. 260)
- 117. a Longitudinal acceleration feedback is the means of reconciling the two demands. (L3, PDF p. 262)
- 118. b The acceleration bias pulls the throttles back before the speed is reached. (L3, PDF p. 262)
- 119. c The acceleration bias cancels the false speed rise so the levers do not move. (L3, PDF p. 262)
- 120. c The loop would read the gust as excess speed and close the throttles. (L3, PDF p. 262)

Mode overview and yaw damping

PDF pages 266–270 · 21 questions

LEVEL 1 — RECALL

121. On a jet aircraft the swept wing lowers which stability with respect to the lateral stability ●
- The directional static stability margin
 - The vertical dynamic stability
 - The longitudinal dynamic stability
122. The oscillation caused by longitudinal dynamic instability takes place about which axes ●
- The yaw and pitch axes
 - The pitch and roll axes
 - The yaw and roll axes
123. The angular velocity about the yaw axis is sensed by which device ●
- An accelerometer
 - A rate gyro
 - A pitot probe
124. The band pass filter in the yaw damper suppresses which unwanted signals ●
- Speed adaption signals from the air data computer
 - Rudder pedal movements made by the pilot
 - Structural oscillations sensed by the rate gyro
125. An all weather operation approval may be restricted in which two ways ●
- In scope and in time
 - In crew and in aircraft
 - In altitude and in speed

LEVEL 2 — UNDERSTAND AND EXPLAIN

126. Insufficient longitudinal dynamic stability spoils which two things ●
- Controllability of the aircraft and passenger comfort
 - Accuracy of the ILS and of the automatic landing system
 - Range of the aircraft and the life of the engines
127. A jet is rolling and slipping sideways in a repeating four second cycle — what is this motion called? ●
- Dutch roll
 - Phugoid
 - Spiral dive
128. A damper is fitted to stop the roll and slip motion of a swept wing jet — which surface does it move? ●
- The rudder
 - The ailerons
 - The spoilers
129. A yaw damper is required during which part of the flight ●
- The flare
 - The take-off roll
 - The enroute phase

- 130.** The yaw rate signal is adapted to airspeed in which manner ●
- Directly proportional to the airspeed
 - Inversely proportional to airspeed
 - Independently of the airspeed
- 131.** A constant yaw rate signal reaching the band pass filter is treated in which way ●
- It is amplified
 - It is passed unchanged
 - It is damped out
- 132.** The yaw damper actuator output is combined with which other demand at the mechanical summing point
- The rudder pedal input
 - The speed adaption signal
 - The band pass filter output
- 133.** The yaw damper is making continuous rudder corrections in the cruise — what do the rudder pedals do? ●
- They move in step with the rudder
 - They are locked until the damper drops out
 - Nothing, they stay where they are
- 134.** An operator intends to fly in all weather conditions — which flight rules must its approval take in? ●
- Instrument flight rules
 - Special visual flight rules
 - Visual flight rules
- 135.** An operator gains an all weather approval in its air operators certificate — what does that approval cover? ●
- Each individual flight crew member
 - The aircraft type or series of types
 - Each individual aircraft registration
- 136.** The all weather approval holds good for which runways ●
- Only those listed in the operator's own route manual
 - Those approved by the authority for that category
 - Only those fitted with a roll out guidance system
- 137.** Directional guidance for take-off in low visibility is covered by which subpart of the all weather operations requirements
- Subpart 2
 - Subpart 4
 - Subpart 1

LEVEL 3 — APPLY AND REASON

- 138.** Dutch roll grows into a problem at high altitude for which reason ●
- The lift of the swept wing rises with altitude
 - The rudder travel available falls off with altitude
 - The damping aerodynamic forces fall with altitude

- 139.** The yaw damper becomes unserviceable during the cruise — what must then be done with the aircraft? ●
- It has to climb to a higher altitude
 - It has to descend to a lower altitude
 - It has to slow to the minimum airspeed
- 140.** The aircraft accelerates in the cruise — what happens to the yaw rate signal passed on to the actuator? ●
- It is cut back as the speed rises
 - It is built up as the speed rises
 - It is held at its take-off value
- 141.** The pilot presses a rudder pedal while the yaw damper is working — how do the two demands reach the rudder?
- They are added or subtracted at a summing point
 - The pedal demand switches the damper out of use
 - The damper demand overrides any pedal demand made

Answers — Mode overview and yaw damping

- 121. c** The swept wing costs longitudinal dynamic stability relative to the lateral. (L1, PDF p. 266)
- 122. c** The motion is a combined yaw and roll oscillation. (L1, PDF p. 266)
- 123. b** Yaw rate is picked up by a rate gyro. (L1, PDF p. 268)
- 124. c** Airframe structural modes sensed by the gyro are filtered out. (L1, PDF p. 268)
- 125. a** Scope and time are the two named restrictions. (L1, PDF p. 270)
- 126. a** Both handling and passenger comfort suffer. (L2, PDF p. 266)
- 127. a** The combined roll and sideslip oscillation is named Dutch roll. (L2, PDF p. 266)
- 128. a** The yaw damper works through the rudder. (L2, PDF p. 266)
- 129. c** It is the cruise, at altitude, where the damper is needed. (L2, PDF p. 266)
- 130. b** The adaptation is inversely proportional to speed. (L2, PDF p. 268)
- 131. c** A steady yaw rate is damped by the filter rather than fed through. (L2, PDF p. 268)
- 132. a** Damper demand and pedal demand are summed before the rudder actuator. (L2, PDF p. 268)
- 133. c** In series mode a damper input does not move the pedals. (L2, PDF p. 268)
- 134. a** The approval must cover IFR flight before all weather operation can follow. (L2, PDF p. 270)
- 135. b** The approval is granted against a type or series of types. (L2, PDF p. 270)
- 136. b** Approval follows the authority's own runway classification for the category. (L2, PDF p. 270)
- 137. b** Subpart 4 deals with low visibility take-off guidance. (L2, PDF p. 270)
- 138. c** Thin air means weaker aerodynamic damping, so the oscillation is no longer suppressed naturally. (L3, PDF p. 266)
- 139. b** Lower down the natural aerodynamic damping is strong enough. (L3, PDF p. 266)
- 140. a** Inverse proportionality means the signal is reduced as speed increases. (L3, PDF p. 268)
- 141. a** Both demands are summed, so neither cancels the other. (L3, PDF p. 268)

All weather operation: definitions and RVR

PDF pages 272–278 · 30 questions

LEVEL 1 — RECALL

142. ILS precision approaches are subdivided into which categories ●
- a. CAT A to CAT C
 - b. CAT I to CAT III
 - c. CAT I to CAT VIII
143. A low visibility take-off is any take-off made below which runway visual range ●
- a. 400 metres
 - b. 300 metres
 - c. 550 metres
144. The decision height is measured above which reference ●
- a. The mean sea level datum
 - b. The runway threshold
 - c. The touch down zone
145. The alert height is defined by which party ●
- a. The airport operator
 - b. The aircraft manufacturer
 - c. The national aviation authority
146. A system failure during an approach can be met by which three courses of action ●
- a. Revert, divert or go straight in
 - b. Continue, descend or go straight in
 - c. Continue, revert or go around

LEVEL 2 — UNDERSTAND AND EXPLAIN

147. The reported runway visual range for departure is 350 metres — how is that take-off classified? ●
- a. As an ordinary take-off
 - b. As a CAT II category take-off
 - c. As a low visibility take-off
148. A CAT III b approach is being planned — what decision height condition applies? ●
- a. No decision height and no visibility limit
 - b. Below 100 ft but not below 50 ft
 - c. Below 50 ft or no decision height
149. An airport has no runway visual range measuring equipment — which value may be converted instead? ●
- a. The decision height selected
 - b. The meteorological visibility
 - c. The cloud base height measured
150. The aircraft reaches the decision height without the required visual reference — what must the crew do? ●
- a. Continue down to the alert height
 - b. Revert to a lower category
 - c. Initiate a missed approach

- 151.** The alert height is quoted in an aircraft's approach data — which approaches does it belong to? ●
- CAT I approaches flown with the autopilot engaged
 - CAT III approaches with a fail-operational autoland
 - CAT II approaches with a fail-passive autoland fitted
- 152.** The value chosen for the alert height depends on what ●
- The failure probability of the autoland
 - The reported runway visual range of the day
 - The landing distance available on the runway
- 153.** The runway visual range refers to a pilot placed where ●
- Over the threshold on the glide path
 - At the holding point beside the runway
 - On the centre line of the runway
- 154.** The minimum runway visual range required for a CAT III approach depends on which two things ●
- The equipment carried and the autoland capability
 - The decision height and the alert height selected
 - The runway length and the approach lighting fitted
- 155.** The crew needs to know the capability of their automatic landing system — where do they read it? ●
- As an amber light on the ILS deviation alert
 - As an approach category on the flight annunciator
 - As a decision height index on the radio altimeter
- 156.** The crew decides to continue the planned approach after a failure — how far down may they go? ●
- The alert height of the automatic landing system
 - The touch down zone marking on the runway
 - Its decision height or minimum descend altitude
- 157.** A failure leaves the aircraft able to fly a lesser approach — what does reverting mean here? ●
- Changing to a category the runway lighting supports
 - Changing to a category the remaining equipment allows
 - Changing to a category the reported weather would allow
- 158.** A go around after a system fault is flown in order to do what ●
- Reduce the landing weight before a second attempt
 - Investigate the failure and plan another approach
 - Allow the ground equipment to be reset by control
- 159.** Besides the type of failure, what else decides the action to be taken
- The present position of the aircraft on approach
 - The category of runway lighting then in use
 - The age of the autoflight equipment then installed

LEVEL 3 — APPLY AND REASON

- 160.** An approach is planned to a decision height of 150 ft with a reported visual range of 400 m — which category applies? ●
- CAT II
 - CAT III a
 - CAT I

161. The reported runway visual range is 100 m and the decision height is 40 ft — which category covers it? ●
- CAT III a
 - CAT III c
 - CAT III b
162. A failure appears in the redundant part of the autoland system above the alert height — what follows? ●
- The approach is broken off and a go around flown
 - The approach reverts to the next higher category
 - The failure is ignored and the approach continued
163. Required ground equipment fails at a point below the alert height — what is the consequence? ●
- A go around is initiated without any delay
 - The failure is ignored and the approach goes on
 - The approach reverts to a higher category minimum
164. A CAT III A approach with a decision height below 100 ft is flown with a fail-passive landing system — what is the minimum visual range?
- 125 m
 - 200 m
 - 75 m
165. The automatic landing system is fail-passive and the decision height is below 50 ft — what visual range may be flown?
- It is not authorised at all
 - It is authorised down to 75 metres
 - It is authorised down to 125 metres
166. A CAT III B approach to a decision height below 50 ft with a fail-operational system and a fail-passive roll-out system needs which minimum visual range
- 75 m
 - 200 m
 - 125 m
167. An approach flown with no decision height is authorised only where the roll-out system holds which standard
- Fail-safe
 - Fail-operational
 - Fail-passive
168. Two aircraft have a fail-operational landing system, one with a roll-out system and one without, both at a decision height below 50 ft — how do their minima differ?
- Only the one without roll-out may fly the approach
 - Only the one with roll-out may fly the approach
 - Both may fly it to the same visual range
169. Reversion to another approach category may be considered only where the failure happens in which band ●
- Above 1000 ft above the ground
 - At or below the alert height
 - Below 1000 ft above the ground

- 170.** A fail-operational autoflight system fails below 1000 ft but above the alert height — what action follows? ●
- The approach continues to decision height
 - A reversion to CAT I is flown
 - A go around is executed
- 171.** After a failure the weather allows a higher minimum and enough autoflight capability remains — what may the crew then do? ●
- Descend below the alert height
 - Continue down to the original minimum
 - Fly that approach category instead

Answers — All weather operation: definitions and RVR

- 142. b** The subdivision runs from CAT I to CAT III. (L1, PDF p. 272)
- 143. a** Below 400 m RVR the departure counts as a low visibility take-off. (L1, PDF p. 272)
- 144. b** Decision height is referenced to the runway threshold. (L1, PDF p. 274)
- 145. b** It is the manufacturer who sets the alert height. (L1, PDF p. 274)
- 146. c** Those three are the courses of action named. (L1, PDF p. 278)
- 147. c** 350 m is below the 400 m threshold, so the take-off is a LVTO. (L2, PDF p. 272)
- 148. c** CAT III b covers decision heights under 50 ft and the no decision height case. (L2, PDF p. 272)
- 149. b** Met visibility can be converted where no RVR equipment exists. (L2, PDF p. 272)
- 150. c** Without the visual reference at decision height a missed approach is compulsory. (L2, PDF p. 274)
- 151. b** The term belongs to fail-operational CAT III approaches. (L2, PDF p. 274)
- 152. a** The page ties the alert height to one thing only: the failure probability of the autoland system. (L2, PDF p. 274)
- 153. c** The definition places the pilot on the runway centre line. (L2, PDF p. 274)
- 154. a** Installed equipment and autoland capability together fix the minimum RVR. (L2, PDF p. 276)
- 155. b** Capability shows up as the available approach category on the flight mode annunciator. (L2, PDF p. 276)
- 156. c** Continue means fly the planned approach to its own minimum. (L2, PDF p. 278)
- 157. b** Reversion is driven by what the surviving equipment can still do. (L2, PDF p. 278)
- 158. b** The go around buys time to look at the failure and reconfigure. (L2, PDF p. 278)
- 159. a** Where the aircraft is on the approach matters as much as what failed. (L2, PDF p. 278)
- 160. a** A decision height between 200 ft and 100 ft with at least 300 m visual range is CAT II. (L3, PDF p. 272)
- 161. c** A decision height below 50 ft with a visual range between 200 m and 75 m falls in CAT III b. (L3, PDF p. 272)
- 162. a** Above the alert height such a failure forces a go around. (L3, PDF p. 274)
- 163. b** At or below the alert height the failure is ignored. (L3, PDF p. 274)
- 164. b** Every CAT III A column in the table shows 200 m. (L3, PDF p. 277)
- 165. a** In the fail-passive column of the table the row for a decision height below 50 ft reads 'Not authorized'. (L3, PDF p. 277)
- 166. c** Under the fail-operational landing system, the column for a fail-passive roll-out system gives 125 m at a decision height below 50 ft. (L3, PDF p. 277)
- 167. b** In the no decision height row only the fail-operational roll-out column carries a value; the fail-passive roll-out column reads 'Not authorized'. (L3, PDF p. 277)
- 168. b** Without a roll-out system that decision height is not authorised at all. (L3, PDF p. 277)
- 169. a** The general rule puts the reversion decision above 1000 ft AGL. (L3, PDF p. 278)
- 170. c** In that band any failure of a fail-operational system means a go around. (L3, PDF p. 278)
- 171. c** Both conditions together allow the higher minimum approach to be flown. (L3, PDF p. 278)

Objectives, safety and CAT III requirements

PDF pages 280–286 · 31 questions

LEVEL 1 — RECALL

- 172.** The objective of CAT II autoflight operation is to hold which level of safety ●
- a. One set by the aircraft manufacturer alone
 - b. One equivalent to manual flight operation
 - c. One above that of manual flight operation
- 173.** In a fail-safe system the output signal after a failure takes which state ●
- a. An arbitrary but safe state
 - b. The last valid state before failure
 - c. A defined and equally safe state
- 174.** In a fail-passive system the output signal once the system has failed takes which state ●
- a. A state within acceptable limits
 - b. A well defined failure state
 - c. An arbitrary but safe state

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 175.** The CAT II safety level depends partly on what the crew can see — which outside aid is named? ●
- a. Approach and runway lighting
 - b. Ground radar surveillance cover
 - c. Runway friction measurement
- 176.** The aircraft reaches the decision height on a CAT II approach — what is the position regarding a manual landing? ●
- a. It is allowed but is not compulsory
 - b. It is forbidden in every case
 - c. It is compulsory in every single case
- 177.** CAT III autoflight operation holds a safety level equivalent to manual flight under which conditions ●
- a. The worst weather and visibility
 - b. Normal weather and visibility conditions
 - c. Conditions no worse than CAT II minima
- 178.** From a CAT III decision height a manual landing is possible to what extent ●
- a. It is compulsory for the pilot flying
 - b. It is possible but is not compulsory
 - c. It is normally not possible at all
- 179.** An aircraft is to be certified for CAT III operation — which equipment is compulsory? ●
- a. An automatic or hybrid landing system
 - b. A flight director approach coupler
 - c. A head-up ground roll guidance display
- 180.** How does the reliability demanded of CAT III systems compare with that demanded for CAT II? ●
- a. It is somewhat lower still
 - b. It is considerably higher
 - c. It is very much the same

- 181.** A designer cuts the number of components in a system — why does availability improve?
- Fewer components leave a faster response time
 - Fewer components leave fewer ways to fail
 - Fewer components leave a lower system weight
- 182.** For CAT III operation the regulations demand redundant systems showing what ●
- Automatic reset after every detected failure
 - Identical hardware from a single manufacturer
 - Defined behaviour for any conceivable failure
- 183.** To behave passively when a fault appears a system needs which provision during normal operation ●
- Periodic self testing carried out at every engine start
 - Manual selection of the standby channel by the crew
 - Permanent monitoring with suppression of any failure
- 184.** A system that stays operative after a failure is described by which term ●
- Failure suppressed
 - Failure tolerant
 - Failure neutral
- 185.** With passive redundancy a drop out of the main system causes what ●
- A manual selection of the remaining system
 - An automatic shut down of all parallel systems
 - An automatic change to the redundant system
- 186.** In passive redundancy the system held in reserve carries which name ●
- The hot standby system
 - The monitored main system
 - The active parallel system
- 187.** A main channel drops out and a reserve takes over automatically — what can be said of the system function afterwards? ●
- The complete functionality is restored
 - Only the monitored functions are restored
 - Only half the functionality is restored
- 188.** For decision heights below 100 ft the two ILS receivers must feed the pilot stations in which way
- Each pilot station from its own receiver
 - Both pilot stations from the same receiver
 - Both receivers into one combined display
- 189.** A decision height of 50 ft or greater is planned — what standard must the automatic landing system hold? ●
- Fail-operational
 - Fail-safe
 - Fail-passive
- 190.** A decision height below 50 ft is planned — what standard must the automatic go-around system hold? ●
- Fail-safe
 - Fail-operational
 - Fail-passive

- 191.** For a no decision height approach the automatic ground roll control may hold which standards ●
- a. Fail-operational or fail-passive
 - b. Fail-passive or fail-safe
 - c. Fail-operational or fail-safe
- 192.** An approach with no decision height is to be flown — what provision is required for speed control? ●
- a. Head-up speed guidance
 - b. Manual throttle control only
 - c. Automatic throttle control

LEVEL 3 — APPLY AND REASON

- 193.** A CAT III A decision height is set so that the crew can judge which thing ●
- a. Whether the autoland has dropped to fail-passive
 - b. Whether the roll out can be supervised visually throughout
 - c. Whether touch down will fall in the touch down zone
- 194.** A CAT III B minimum is set so as to secure which crew capability ●
- a. Visual confirmation of the touch down point
 - b. Visual acquisition of the approach light system
 - c. Visual supervision of the roll out
- 195.** Automatic guidance of the roll out until taxi speed is reached is demanded for which categories ●
- a. CAT III B and CAT III C
 - b. CAT III A and CAT III B
 - c. CAT II and CAT III A
- 196.** One of several parallel channels drops out and the output signal is unaffected — which redundancy is in use? ●
- a. Monitored redundancy
 - b. Active redundancy
 - c. Passive redundancy
- 197.** An operator wants to certify a CAT II aircraft with no automatic throttle system — what must be shown?
- a. That the engines accelerate within the regulated time
 - b. That speed can be held by hand without undue workload
 - c. That the autopilot can hold speed within ten knots
- 198.** The number of ILS receivers and radio altimeters may have to be raised for which reason
- a. To give an alert height indication at both stations
 - b. To give fail-operational capability where required
 - c. To give a roll out guidance display at both stations
- 199.** The decision height planned is below 50 ft — what standard must the automatic landing system then meet? ●
- a. Fail-safe
 - b. Fail-passive
 - c. Fail-operational

- 200.** Which wheel provision is required for an approach flown with no decision height ●
- a. An anti-skid braking system
 - b. An autobrake disarm warning
 - c. A carbon brake wear indicator
- 201.** A no decision height approach is planned — what standard must the automatic landing system and the go-around system meet? ●
- a. Fail-operational landing with fail-passive go-around
 - b. Fail-operational landing with fail-operational go-around
 - c. Fail-passive landing with fail-operational go-around
- 202.** A fail-passive approach system without automatic landing is acceptable at 50 ft provided the aeroplane holds which property
- a. A low approach speed and good manoeuvrability
 - b. A high approach speed and a long wheel base
 - c. A fail-operational go-around capability fitted

Answers — Objectives, safety and CAT III requirements

- 172. b** The aim is equivalence with manual operation, under worse weather. (L1, PDF p. 280)
- 173. a** Fail-safe gives an arbitrary state that is nevertheless safe. (L1, PDF p. 283)
- 174. b** Fail-passive gives a defined, known failure state. (L1, PDF p. 283)
- 175. a** Outside visual cues from the approach and runway lighting are listed. (L2, PDF p. 280)
- 176. a** CAT II minima permit a manual landing without requiring one. (L2, PDF p. 280)
- 177. a** CAT III carries that equivalence into the worst conditions. (L2, PDF p. 280)
- 178. c** Unlike the lower categories, CAT III leaves no room for a manual landing from the decision height. (L2, PDF p. 280)
- 179. a** CAT III requires an automatic or hybrid landing system. (L2, PDF p. 280)
- 180. b** CAT III calls for considerably higher reliability than CAT II. (L2, PDF p. 280)
- 181. b** Less complexity means fewer ways for the system to fail. (L2, PDF p. 283)
- 182. c** The behaviour at every conceivable failure must be defined. (L2, PDF p. 283)
- 183. c** Continuous monitoring plus suppression is what makes passive behaviour possible. (L2, PDF p. 283)
- 184. b** Such a system is called failure tolerant. (L2, PDF p. 283)
- 185. c** The change over happens automatically. (L2, PDF p. 283)
- 186. a** The reserve system is called the hot standby. (L2, PDF p. 283)
- 187. a** The standby restores the full function after the change over. (L2, PDF p. 283)
- 188. a** The two stations must be fed from separate receivers. (L2, PDF p. 286)
- 189. c** At 50 ft and above a fail-passive landing system is enough. (L2, PDF p. 286)
- 190. c** The go-around system is required to be fail-passive there. (L2, PDF p. 286)
- 191. a** Either standard is accepted for the ground roll function. (L2, PDF p. 286)
- 192. c** Automatic throttle control is on the no decision height equipment list. (L2, PDF p. 286)
- 193. c** At CAT III A the crew can still see whether the aircraft will land in the zone. (L3, PDF p. 280)
- 194. c** The CAT III B minimum exists so the roll out can be watched. (L3, PDF p. 280)
- 195. a** Flare and roll out guidance to taxi speed is listed for CAT III B and C. (L3, PDF p. 280)
- 196. b** Only active redundancy keeps the output going with no switch over. (L3, PDF p. 283)
- 197. b** The way out is an in-flight demonstration of manual speed control. (L3, PDF p. 284)
- 198. b** Extra sensors are what makes a fail-operational installation possible. (L3, PDF p. 286)
- 199. c** Below 50 ft only a fail-operational landing or hybrid system will do. (L3, PDF p. 286)

- 200. a** An anti-skid braking system is one of the items listed for no decision height. (L3, PDF p. 286)
- 201. a** The landing system must be fail-operational while the go-around stays fail-passive. (L3, PDF p. 286)
- 202. a** Those handling properties are what make the manual landing acceptable. (L3, PDF p. 286)

Fail safe and fail passive systems

PDF pages 288–294 · 29 questions

LEVEL 1 — RECALL

- 203.** Which system is defined as one comprising an automatic and/or a hybrid landing system ●
- a. the guidance system
 - b. the landing system
 - c. the flight control system
- 204.** Which equipment the term landing system covers ●
- a. onboard equipment only
 - b. ground equipment only
 - c. onboard and ground equipment
- 205.** Which failure of a control surface can occur on a fail safe flight control system ●
- a. a slow runaway of the control surface
 - b. a hard over of the control surface
 - c. a complete loss of control surface authority
- 206.** The autopilot runs a surface hard over while the aircraft is enroute. Within which time must the pilot be able to take over control? ●
- a. 3 seconds
 - b. 10 seconds
 - c. 1 second
- 207.** Which kind of redundancy the parallel operation of two fail safe autopilots shows ●
- a. standby redundancy
 - b. active redundancy
 - c. passive redundancy
- 208.** Which arrangement inside one and the same autopilot gives fail passive behaviour ●
- a. three actuators driven by a single channel
 - b. two independent channels working in parallel
 - c. two autopilots coupled rigidly together
- 209.** What the autopilot does when the comparison of its two channels detects a failure ●
- a. it limits its authority
 - b. it reverts to one channel
 - c. it disengages

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 210.** Which items the landing system comprises in accordance with JAR-AWO 321 ●
- a. all sensors, antennas and ground stations
 - b. all instruments, displays and ground transmitters
 - c. all sensors, instruments and power supplies
- 211.** What the differing architectures of the two major transport aircraft manufacturers are meant to realize
- a. a lower weight of the autopilot installation
 - b. a common standard of autopilot engagement
 - c. failure modes acceptable for CAT III operation

- 212.** Two fail passive single systems are operated together in dual operation. Which behaviour does the resulting system show? ●
- fail safe behaviour
 - fail operational behaviour
 - fail passive behaviour
- 213.** Why a hard over is not suppressed by a fail safe flight control system ●
- because it cannot recognize the failure
 - because its servos have no force limiter
 - because its actuators have no clutch
- 214.** How the effect of a failure is limited on a fail safe system
- by reducing the authority of the servos
 - by comparing the two autopilot channels
 - by voting between three actuator outputs
- 215.** Under which condition authority limitation is allowed at all ●
- only if a second autopilot is installed
 - only if the servos carry a torque limiter
 - only if a manual flight control system exists
- 216.** Which effect a failure of a fail passive flight control system has on the trim state and the flight path ●
- neither of them is changed considerably
 - both of them are changed considerably
 - the trim state changes but the flight path does not
- 217.** What can no longer be carried out once a fail passive flight control system has failed ●
- an automatic landing
 - a manual go around
 - an ILS approach
- 218.** The fail passive automatic flight control system fails during an approach. What must happen then? ●
- the system continues the approach and landing by itself
 - the pilot has to take over the control of the aircraft
 - the second autopilot takes over control and lands
- 219.** How the independence of two autopilots working in dual operation is achieved ●
- by one common computer serving both of the channels
 - by a common power supply feeding two sets of sensors
 - by separate supplies, sensors, computers and actuators
- 220.** Which causes make the rigidly coupled actuators of two autopilots work against each other
- different reference values in the two sensor units
 - tolerances, pressure differences or poor adjustment
 - a delay between the two autopilot computers
- 221.** What is done for safety reasons in the two channels of one fail passive autopilot ●
- the same hardware and the same programs in both lanes
 - one of the two channels is left unpowered as a spare
 - different hardware and different programming languages

- 222.** A dual channel autopilot has produced an output in each of its channels. What happens to the two signals before the actuator sees them? ●
- they are added together and then released
 - they are sent on separately to two actuators
 - they are compared for validity and then released
- 223.** By which three means fail passive behaviour is reached in a dual channel autopilot ●
- mechanical voting, torque limiting and authority limiting
 - parallel processing, mutual monitoring and output comparison
 - active redundancy, crosstalk and autopilot synchronisation

LEVEL 3 — APPLY AND REASON

- 224.** An autopilot that detects its own failure and disengages itself is operated on its own. Which behaviour does the system then show? ●
- fail operational behaviour
 - fail passive behaviour
 - fail safe behaviour
- 225.** A manufacturer limits the authority of its autopilot servos, but the aircraft has no manual control path strong enough to overrule the autopilot. Why is that not acceptable? ●
- because the autopilot could then not recognize its own hard over
 - because the servos would then not reach their full travel range
 - because the pilot could then not over-power the autopilot
- 226.** A single autopilot suffers a failure that it cannot recognize at all. Which failure mode is that, and what can the control surface do? ●
- fail passive, and the surface keeps its position
 - fail operational, and the landing is continued
 - fail safe, and a hard over of the surface is possible
- 227.** A fail passive system must be able to disengage itself. How does it recognize that it has a failure? ●
- by comparing the two autopilot computers with each other
 - by monitoring the hydraulic pressure at the actuator
 - by comparing the surface position with the commanded position
- 228.** During dual operation the two autopilot actuators try to deflect one surface in opposite directions. What does the surface do?
- it follows the faster of the two actuators
 - it runs to its full deflection
 - it stays in its position
- 229.** Two fail safe autopilots drive one surface in dual operation. What does the feel system contribute?
- the torque limit applied at the actuator clutch
 - the neutral reference for the deflection limiter
 - the third signal the mid-value voter needs
- 230.** A fail passive autopilot built from two channels sends its signal to only one actuator. What follows for force conflicts? ●
- force conflicts have to be limited by a voter
 - no force conflict can arise
 - a force conflict arises at every input

- 231.** In which respect a dual channel fail passive autopilot differs from two parallel fail safe autopilots ●
- it drives three actuators and needs a mid-value voter
 - it drives two actuators and needs a torque limiter
 - it drives one actuator only and needs no mechanical voting

Answers — Fail safe and fail passive systems

- 203. c** That wording is the definition of the flight control system. (L1, PDF p. 288)
- 204. a** The definition limits the term landing system to equipment carried on board. (L1, PDF p. 288)
- 205. b** A hard over of a control surface is named as the most important feature of the fail safe failure case. (L1, PDF p. 290)
- 206. a** FAR 25 allows 3 seconds for the takeover enroute. (L1, PDF p. 290)
- 207. b** Two fail safe autopilots working in parallel are described as active redundancy. (L1, PDF p. 292)
- 208. b** Fail passive behaviour can come from two independent channels inside one autopilot. (L1, PDF p. 294)
- 209. c** A detected failure disengages the autopilot. (L1, PDF p. 294)
- 210. c** The equipment named in the definition is sensors, instruments and power supplies. (L2, PDF p. 288)
- 211. c** The architectures exist to produce the fail passive or fail operational modes that CAT III operation accepts. (L2, PDF p. 288)
- 212. b** Dual operation of fail passive single systems yields fail operational behaviour. (L2, PDF p. 288)
- 213. a** Suppression is impossible because the system has no way of recognizing the failure. (L2, PDF p. 290)
- 214. a** Reducing servo authority is the means given for limiting the failure effect. (L2, PDF p. 290)
- 215. c** A manual flight control system able to over-power the autopilot is the condition for allowing authority limitation. (L2, PDF p. 290)
- 216. a** A fail passive failure leaves the trim state unchanged and causes no considerable deviation from flight path or attitude. (L2, PDF p. 292)
- 217. a** The printed definition says an automatic landing can no longer be performed. (L2, PDF p. 292)
- 218. b** On failure of a fail passive system the pilot has to take over control. (L2, PDF p. 292)
- 219. c** Independence comes from everything being separate: supplies, sensors, computers, actuators and cable ducts. (L2, PDF p. 292)
- 220. b** Force conflicts are attributed to tolerances, pressure differences and poor adjustment. (L2, PDF p. 292)
- 221. c** The channels differ in hardware and in the programming language of the identical programs. (L2, PDF p. 294)
- 222. c** The outputs are compared for validity, and only then released to the actuator. (L2, PDF p. 294)
- 223. b** Those three together are what makes the dual channel autopilot fail passive. (L2, PDF p. 294)
- 224. b** A single system able to detect and remove its own failure gives fail passive behaviour in single operation. (L3, PDF p. 288)
- 225. c** Authority limitation relies on the pilot being able to apply strong inputs and over-power the autopilot manually. (L3, PDF p. 290)
- 226. c** Inability to recognize a failure is the mark of the fail safe mode, in which a hard over of the control surface can occur. (L3, PDF p. 290)
- 227. c** Recognition rests on the electrical comparison of actual surface position with commanded actuator position. (L3, PDF p. 292)
- 228. c** Opposing actuator commands are voted out, so the surface holds its position. (L3, PDF p. 292)
- 229. c** Mid-value voting needs three inputs, and with two autopilots the feel system provides the missing one. (L3, PDF p. 292)
- 230. b** With a single actuator there is nothing to work against, so no force conflict occurs. (L3, PDF p. 294)
- 231. c** Parallel fail safe autopilots need mechanical voting against force conflicts, while the dual channel autopilot drives a single actuator and does not. (L3, PDF p. 294)

Fail operational and hybrid landing systems

PDF pages 296–302 · 19 questions

LEVEL 1 — RECALL

- 232.** What a fail operational flight control system does at any failure below the alert height ●
- a. it continues the approach and lands automatically
 - b. it disengages and hands the approach to the pilot
 - c. it freezes the control surfaces in their position
- 233.** Of which two parts a hybrid landing system consists ●
- a. a fail safe landing system and a mechanical voter
 - b. a fail operational landing system and a second autopilot
 - c. a fail passive landing system and a guidance system
- 234.** Which systems are usually used as the supporting system of a hybrid landing system ●
- a. radio altimeter systems
 - b. yaw damper systems
 - c. head up guidance systems

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 235.** Which failure mode the automatic landing system has after a failure has occurred ●
- a. fail passive
 - b. fail safe
 - c. fail operational
- 236.** Why mechanical voting has to be applied in triple operation
- a. to detect a hard over of a surface
 - b. to limit the authority of the servos
 - c. to prevent force conflicts
- 237.** Two autopilots run in parallel and each one disengages itself on its own failure. Which kind of redundancy does that architecture show? ●
- a. active redundancy
 - b. passive redundancy
 - c. standby redundancy
- 238.** The main system of a hybrid landing system fails on the approach. What does the supporting guidance system then allow? ●
- a. automatic continuation of approach and landing
 - b. an automatic go around at the alert height
 - c. manual continuation of approach and landing
- 239.** Of which units the hybrid built from fail safe autopilots consists ●
- a. three fail safe autopilots and one head up display
 - b. one fail safe autopilot and two flight director systems
 - c. two fail safe autopilots and one flight director system
- 240.** Of which units the hybrid system built around a fail passive autopilot consists ●
- a. one fail passive autopilot and one flight director system
 - b. two fail passive autopilots and one flight director system
 - c. one fail passive autopilot and two head up displays

LEVEL 3 — APPLY AND REASON

- 241.** A single autopilot fails below the alert height in a fail operational system. What can the crew still expect of the automatic system? ●
- the system disengages and the approach has to be abandoned
 - the approach continues but the landing has to be flown manually
 - approach, flare, landing and roll out are completed automatically
- 242.** A fail operational system has already suffered one failure. What would a second failure mean now? ●
- the landing is still completed, as the system stays fail operational
 - a hard over may occur, as the system is now fail safe
 - the pilot must take over, as the system is now fail passive
- 243.** In a triple operation system one autopilot output drifts away from the other two. What does the mid-value voter do to the surface?
- the surface follows the odd output until that autopilot is off
 - the surface follows the middle actuator and the odd output is lost
 - the surface takes the average, so the odd output shifts it
- 244.** Why two fail passive autopilots in parallel give a fail operational system ●
- because both actuators drive the control surface together
 - because each one detects its own failure and disengages itself
 - because the two outputs are voted mechanically against each other
- 245.** Two fail passive autopilots work in parallel on the same control train. Why can no force conflict arise there? ●
- because only one actuator drives the surface at a time
 - because a mid-value voter selects the middle of the signals
 - because a torque limiter is fitted in each actuator clutch
- 246.** One of two parallel fail passive autopilots detects its own failure during an automatic landing. What happens next? ●
- both of them disengage and the pilot takes over
 - it disengages itself and the other one keeps control
 - the faulty one is voted out by the mechanical voter
- 247.** Of the two fail operational architectures described, one of them needs mechanical voting to avoid force conflicts ●
- triple operation with three fail safe autopilots
 - single operation of one fail passive autopilot
 - parallel operation of two fail passive autopilots
- 248.** Which behaviour the autopilot system of a hybrid built from two fail safe autopilots shows ●
- fail passive, augmented by a flight director
 - fail safe, augmented by an auto-thrust system
 - fail operational, augmented by a flight director
- 249.** The autopilot of a hybrid landing installation fails on short final. How can the landing still be completed? ●
- manually, following the supporting guidance system
 - automatically, after the voter has selected a channel
 - automatically, by the second autopilot taking over

- 250.** Why a hybrid landing system counts as fail operational although its automatic landing system is only fail passive ●
- a. the guidance system allows the landing to be continued manually
 - b. the automatic landing system becomes fail operational below the alert height
 - c. the two autopilots of the system are voted mechanically

Answers — Fail operational and hybrid landing systems

- 232. a** Below the alert height the fail operational system carries on to an automatic landing and roll out. (L1, PDF p. 296)
- 233. c** The main system is a fail passive automatic landing system and it is supported by an independent guidance system. (L1, PDF p. 300)
- 234. c** Head up guidance systems are the usual supporting systems. (L1, PDF p. 300)
- 235. a** The remaining system is fail passive once a failure has happened. (L2, PDF p. 296)
- 236. c** Voting is applied for the same reason as in dual operation: to prevent force conflicts. (L2, PDF p. 296)
- 237. b** This architecture is described as passive redundancy. (L2, PDF p. 298)
- 238. c** The supporting system lets the crew carry on manually after the main system has failed. (L2, PDF p. 300)
- 239. c** It is built from two fail safe autopilots plus at least one flight director system. (L2, PDF p. 300)
- 240. a** This architecture uses a single fail passive autopilot with at least one flight director system. (L2, PDF p. 302)
- 241. c** Fail operational means the whole sequence down to roll out is still flown automatically. (L3, PDF p. 296)
- 242. c** Once fail passive, a further failure means no automatic landing and the pilot has to take over. (L3, PDF p. 296)
- 243. b** Mid-value voting passes only the middle signal, so a single deviating output is suppressed. (L3, PDF p. 297)
- 244. b** Self-detection and self-disengagement leave the healthy autopilot in control, which is fail operational behaviour. (L3, PDF p. 298)
- 245. a** At any instance only one actuator is driving, so nothing can work against it. (L3, PDF p. 298)
- 246. b** The failed autopilot takes itself out and the remaining one continues to control the aircraft. (L3, PDF p. 298)
- 247. a** Three parallel fail safe autopilots drive their actuators together and need voting; two fail passive autopilots never drive together. (L3, PDF p. 298)
- 248. a** Two fail safe autopilots make the autopilot system fail passive, and the flight director augments it. (L3, PDF p. 300)
- 249. a** The hybrid system's reserve is manual flight on the supporting guidance system, not a second automatic system. (L3, PDF p. 300)
- 250. a** The overall capability is kept by the independent guidance system letting the crew finish the landing by hand. (L3, PDF p. 300)

Landing capabilities, autoland warning and equipment

PDF pages 304–310 · 16 questions

LEVEL 1 — RECALL

251. Where the autoland warning is displayed ●
- a. in the glare shield, in the pilot's view
 - b. on the engine display of the lower screen
 - c. on the overhead panel above the pilots
252. Which colour the autoland warning has ●
- a. amber
 - b. red
 - c. white
253. Where components influencing CAT II and CAT III approach capability are specially marked ●
- a. in column 1 of the operations manual part B
 - b. in column 3 of the minimum equipment list
 - c. in chapter 22 of the aircraft maintenance manual

LEVEL 2 — UNDERSTAND AND EXPLAIN

254. The capability logic has checked the serviceability of all required systems. Which supplies does it count as well?
- a. the independent hydraulic and electrical power supplies
 - b. the independent pneumatic and electrical power supplies
 - c. the independent hydraulic and pneumatic power supplies
255. What the degree of redundancy determines ●
- a. the decision height of the approach
 - b. the failure mode of the system
 - c. the alert height of the approach
256. One fail passive autopilot or flight director is operational and the redundancy is limited. Which capability is reached? ●
- a. CAT 3 SINGLE
 - b. CAT 2
 - c. CAT 1
257. Which condition gives a capability of CAT 2 ●
- a. one fail passive autopilot with sufficient redundancy
 - b. two fail passive autopilots with sufficient redundancy
 - c. one fail passive autopilot with limited redundancy
258. The autoland warning comes on during an automatic approach. Which reaction is the normal one? ●
- a. the initiation of a go around
 - b. a continuation of the approach to CAT I minima
 - c. the disengagement of one autopilot

- 259.** The autoland warning comes on and the runway is clearly in sight. What is possible besides a go around? ●
- a. an automatic landing
 - b. a manual landing
 - c. a second automatic approach
- 260.** Above which difference of radio height between two radio altimeters the autoland warning comes on
- a. 100 ft
 - b. 10 ft
 - c. 50 ft

LEVEL 3 — APPLY AND REASON

- 261.** Required ground equipment fails while the aircraft is still above the alert height and the capability changes. What has to be done? ●
- a. a go around has to be initiated
 - b. the autopilot has to be disengaged and the approach flown manually
 - c. the approach may be continued to CAT I minima
- 262.** A single failure occurs below the alert height and degrades the failure mode from fail operational to fail passive. What happens to the displayed capability? ●
- a. it is replaced by a red warning
 - b. it is no longer changed
 - c. it drops by one category
- 263.** A single failure below the alert height degrades the failure mode, yet the capability stays as it was. Why is it kept? ●
- a. because the autoflight system still performs its duty
 - b. because the ground equipment can no longer be monitored
 - c. because the autopilot can no longer be disengaged
- 264.** One fail passive autopilot and one auto-thrust system are operational. Which capability is displayed? ●
- a. CAT 3 SINGLE
 - b. CAT 3 DUAL
 - c. CAT 2
- 265.** Two fail passive autopilots and one auto-thrust system are operational. Which capability is displayed and which failure mode does the system have? ●
- a. CAT 3 SINGLE, and the system is fail passive
 - b. CAT 3 DUAL, and the system is fail operational
 - c. CAT 2, and the system is fail operational
- 266.** Why does the autoland warning demand action at once rather than merely being noted by the crew? ●
- a. because the warning repeats every ten seconds
 - b. because it is a red warning
 - c. because the warning sits on the glare shield

Answers — Landing capabilities, autoland warning and equipment

- 251. a** It sits in the glare shield, inside the pilot's field of view. (L1, PDF p. 306)
- 252. b** The autoland warning is a red warning. (L1, PDF p. 306)

- 253. b** The MEL marks them in its column 3. (L1, PDF p. 310)
- 254. a** The number of independent hydraulic and electrical power supplies is counted as well. (L2, PDF p. 304)
- 255. b** Redundancy decides whether the system is fail safe, fail passive or fail operational. (L2, PDF p. 304)
- 256. c** With one fail passive autopilot or flight director and limited redundancy only CAT I is available. (L2, PDF p. 304)
- 257. a** CAT 2 needs one fail passive autopilot plus sufficient redundancy. (L2, PDF p. 304)
- 258. a** A go around is the normal response to the autoland warning. (L2, PDF p. 306)
- 259. b** With visual conditions permitting, the landing may be completed manually. (L2, PDF p. 306)
- 260. b** The warning condition is a radio height difference of more than 10 ft. (L2, PDF p. 306)
- 261. a** A capability change above the alert height requires a go around. (L3, PDF p. 304)
- 262. b** Below the alert height the capability is frozen even though the failure mode degrades. (L3, PDF p. 304)
- 263. a** The capability stays because the system is still doing its job after the single failure. (L3, PDF p. 304)
- 264. a** That combination gives CAT III A, displayed as CAT 3 SINGLE. (L3, PDF p. 304)
- 265. b** Two fail passive autopilots make the total system fail operational and give CAT 3 DUAL for CAT III B. (L3, PDF p. 304)
- 266. b** Its red colour is what classes it as requiring immediate action. (L3, PDF p. 306)

Helicopter stability augmentation

PDF pages 312–320 · 45 questions

LEVEL 1 — RECALL

267. What the collective pitch lever controls ●
- a. the main rotor lift
 - b. the velocity vector
 - c. the tail rotor thrust
268. What the cyclic pitch stick controls ●
- a. the main rotor lift
 - b. the velocity vector
 - c. the tail rotor thrust
269. The cyclic stick is moved forward. Which movement of the helicopter results? ●
- a. a tail down movement
 - b. a nose down movement
 - c. a left sideward movement
270. The cyclic stick is moved rearward. Which movement of the helicopter results? ●
- a. a right sideward movement
 - b. a nose down movement
 - c. a tail down movement
271. Through which servo the pedals act ●
- a. the front servo
 - b. the LH servo
 - c. the yaw servo
272. What a twist grip throttle is ●
- a. a rotary trim controller at the cyclic pitch stick
 - b. a rotary rpm controller at the collective pitch lever
 - c. a friction lock fitted at the pedal unit
273. On which helicopters a twist grip throttle is found ●
- a. on those powered by a piston engine
 - b. on those powered by a turbine engine
 - c. on those with a counter clockwise rotor
274. Which stability a helicopter has around its longitudinal, lateral and vertical axes ●
- a. positive stability around the longitudinal axis only
 - b. no positive stability around any of them
 - c. positive stability around the vertical axis only
275. Which sensors a helicopter stabilization system uses ●
- a. rate gyros
 - b. accelerometers
 - c. pressure transducers

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 276.** The collective pitch lever is raised. How is the pitch angle of the main rotor blades changed? ●
- on all blades at once and by the same amount
 - on half of the blades at once and by the same amount
 - on each blade in turn and by a different amount
- 277.** How the cyclic stick produces a variation of the rotor blade pitch angle ●
- by changing the pitch of all the blades equally
 - by tilting the main rotor plane around the X or Y axis
 - by changing the thrust produced by the tail rotor
- 278.** Only the front servo is driven by a control input. Around which axis is the rotor plane tilted? ●
- the vertical axis, Z-Z'
 - the lateral axis, Y-Y'
 - the longitudinal axis, X-X'
- 279.** Which servos tilt the rotor plane around the longitudinal axis ●
- the front and yaw servos
 - the front and LH servos
 - the LH and RH servos
- 280.** What the up and down movement of the swash plate is called ●
- lateral pitch variation
 - collective pitch variation
 - cyclic pitch variation
- 281.** The swash plate is moved up from its initial position. Which quantity at the rotor changes? ●
- the angle of attack of the rotor blades
 - the length of the blade lever rods
 - the rotational speed of the main rotor
- 282.** What decides about which axis the swash plate is tilted ●
- the position of the two pedal units
 - the movement direction of the cyclic stick
 - the amount of collective pitch applied
- 283.** The swash plate is to be tilted about the lateral axis. Which servo is energized on its own? ●
- the LH servo
 - the front servo
 - the yaw servo
- 284.** A sideward movement is commanded. How are the LH and RH servos driven for it? ●
- concurrently, with equal signals in opposite directions
 - one after the other, with signals of different magnitude
 - concurrently, with equal signals in the same direction
- 285.** Which movement of the swash plate a diagonal cyclic stick movement produces ●
- an up and down movement without any tilt
 - a combined tilt about both of the axes
 - a tilt about the lateral axis only

- 286.** The pilot presses one of the pedals. What is changed at the tail rotor? ●
- a. the rotational speed of the tail rotor
 - b. the diameter of the tail rotor disc
 - c. the pitch angle of the tail rotor blades
- 287.** Around which axis the tail rotor allows the helicopter to be controlled ●
- a. the vertical axis
 - b. the lateral axis
 - c. the longitudinal axis
- 288.** What gives an aeroplane the stability a helicopter lacks ●
- a. the rate gyros feeding its stabilization computer
 - b. the dihedral and sweep of the wing and the stabilizers
 - c. the attenuation signals sent to its control servos
- 289.** Why stabilization systems have been developed for helicopters ●
- a. to allow the main rotor to be driven at a lower speed
 - b. to let the pilot concentrate on navigation and supervision
 - c. to raise the maximum forward speed of the helicopter
- 290.** The SCAS has received a rate signal from one of the gyros. What does it drive with its attenuation signal? ●
- a. the servo valves of the control servos
 - b. the main servo input levers directly
 - c. the trim actuators of the cyclic stick
- 291.** Which stability the helicopter shows once the stabilization system is working ●
- a. neutral stability
 - b. positive stability
 - c. negative stability
- 292.** What holds the cable system of the main rotor controls in position ●
- a. the control servos
 - b. the motion transducers
 - c. the trim actuators
- 293.** The control servos of a stabilized helicopter are driven by the SCAS computer. What do they move? ●
- a. the swash plate directly
 - b. the main servo input levers
 - c. the trim control panel switches
- 294.** The pilot returns the cyclic stick to its neutral position. What does the output of the motion transducer do? ●
- a. it reaches its maximum
 - b. it becomes zero
 - c. it reverses its sign
- 295.** From which sources the stored reference value is obtained ●
- a. from the motion transducers, via the trim actuator units
 - b. from the rate gyros, via the stability augmentation computer
 - c. from the attitude and heading gyros, via the synchronizer

- 296.** To which fixed wing function this way of controlling the helicopter is comparable ●
- to the yaw damper
 - to control wheel steering
 - to the autothrottle
- 297.** What is compared with each other in the flight path control of a helicopter
- the trim signals with the outputs of the motion transducers
 - the commanded path with the signals of the three rate gyros
 - the planned path and altitude with the momentary path parameters
- 298.** A flight path error has been worked out on a helicopter flying an upper mode. Where is it superimposed on the stabilization signals? ●
- in the SCAS computer
 - in the synchronizer control unit
 - in the trim control panel
- 299.** Through which item each trim signal reaches its trim actuator ●
- the respective motion transducer
 - the synchronizer control unit
 - the stability augmentation computer

LEVEL 3 — APPLY AND REASON

- 300.** A helicopter is to be moved along its velocity vector without any change of lift. Which control is used and what does it do to the rotor plane? ●
- the cyclic stick, which raises the whole rotor plane
 - the cyclic stick, which tilts the rotor plane
 - the collective lever, which tilts the rotor plane
- 301.** The cyclic stick is moved to the left-hand side. Which servo stays unenergized and which movement results? ●
- the front servo, and the helicopter moves left sideward
 - the front servo, and the helicopter moves nose down
 - the LH servo, and the helicopter moves left sideward
- 302.** The helicopter is to be moved sideward to the right. Around which axis must the swash plate tilt, and which units do it? ●
- the longitudinal axis, by the front servo alone
 - the lateral axis, by the front servo alone
 - the longitudinal axis, by the LH and RH servos
- 303.** The pedal position is changed in flight. Which quantity at the tail rotor changes as a result of the blade pitch change? ●
- the torque of the main rotor
 - the thrust of the tail rotor
 - the speed of the tail rotor
- 304.** With speed already established the cyclic stick is moved back. Which pair of reactions does that give?
- a climb like an airplane, with speed being gained
 - a climb like an airplane, with speed being lost
 - a descent like an airplane, with speed being lost

- 305.** With speed established the left pedal is applied on such a helicopter. Which pair of reactions results?
- a yaw to the left, with a little speed increase
 - a yaw to the right, with a little speed drop
 - a yaw to the left, with a little speed drop
- 306.** What follows from the helicopter having no positive stability ●
- its rotor speed has to be controlled permanently by the pilot
 - its attitude has to be controlled permanently by the pilot
 - its trim actuators have to run permanently in flight
- 307.** How the pilot gets a feedback of the attitude change at his controls ●
- through the cable system, driven by the main servos
 - through the rate gyros, driven by the SCAS computer
 - through the trim actuators, driven by the control panel
- 308.** The cyclic stick is returned to neutral after a manoeuvre. What does the SCAS computer do with the attitude reached at that moment? ●
- it passes it to the trim actuators as a trim command
 - it stores it as the reference value to be maintained
 - it compares it with the planned flight path of the mission
- 309.** A helicopter flying with its stabilization system engaged is disturbed in roll. How is that disturbance countered? ●
- the trim actuator returns the cyclic stick to its neutral position
 - the motion transducer stores a new attitude as the reference
 - the sensed rate drives the control servos to compensate the motion
- 310.** A pilot input and the stabilization signals are present at the same time. How does the SCAS computer handle them? ●
- it passes only the larger of the two signals to the servos
 - it overlays the attitude reference and stabilization signals
 - it blocks the stabilization signals while the pilot inputs
- 311.** A helicopter loses its stability augmentation in flight. Which quality does it lose, and what does the pilot have to do? ●
- positive stability, and he must retrim the controls permanently
 - neutral stability, and he must hold the rotor speed permanently
 - neutral stability, and he must control the attitude permanently

Answers — Helicopter stability augmentation

- 267. a** The collective pitch lever is the lift control. (L1, PDF p. 312)
- 268. b** The cyclic stick controls the velocity vector. (L1, PDF p. 312)
- 269. b** Forward cyclic gives a nose down movement. (L1, PDF p. 314)
- 270. c** Rearward cyclic gives a tail down movement. (L1, PDF p. 314)
- 271. c** Pedal inputs pass through the yaw servo. (L1, PDF p. 316)
- 272. b** It is the rotary rpm control sitting on the collective pitch lever. (L1, PDF p. 318)
- 273. a** The twist grip throttle belongs to piston engined helicopters. (L1, PDF p. 318)
- 274. b** The helicopter has no positive stability about any of the three axes. (L1, PDF p. 320)
- 275. a** Rotation rates about pitch, roll and yaw are sensed by rate gyros. (L1, PDF p. 320)
- 276. a** Collective input acts on every blade simultaneously and equally. (L2, PDF p. 312)

- 277. b** The cyclic tilts the rotor plane about the X or Y axis. (L2, PDF p. 312)
- 278. b** The front servo tilts the rotor plane about the lateral Y-Y' axis. (L2, PDF p. 312)
- 279. c** The LH and RH servos work the longitudinal X-X' axis. (L2, PDF p. 312)
- 280. b** That up and down movement is the collective pitch variation. (L2, PDF p. 314)
- 281. a** Moving the swash plate up and down alters the blades' angle of attack. (L2, PDF p. 314)
- 282. b** The tilt axis follows the direction in which the cyclic stick is moved. (L2, PDF p. 314)
- 283. b** Only the front servo is energized for a tilt about the lateral axis. (L2, PDF p. 314)
- 284. a** They get equal signals at the same time, but run opposite to each other. (L2, PDF p. 314)
- 285. b** A diagonal input tilts the swash plate proportionally about both axes at once. (L2, PDF p. 314)
- 286. c** The pedals set tail rotor blade pitch. (L2, PDF p. 316)
- 287. a** Tail rotor thrust gives control about the vertical axis, that is yaw control. (L2, PDF p. 316)
- 288. b** Wing dihedral and sweep together with the stabilizers give the fixed wing aircraft its stability. (L2, PDF p. 320)
- 289. b** They relieve the pilot of the constant control task so he can attend to navigation and supervision. (L2, PDF p. 320)
- 290. a** The attenuation signals go to the servo valves of the control servos. (L2, PDF p. 320)
- 291. a** With the stabilization system active the helicopter exhibits neutral stability. (L2, PDF p. 320)
- 292. a** The control servos hold that cable system in position. (L2, PDF p. 320)
- 293. b** The control servos act on the main servo input levers. (L2, PDF p. 320)
- 294. b** At neutral the motion transducer output is zero. (L2, PDF p. 320)
- 295. c** It comes from the pitch-roll attitude gyro and the heading gyro through the synchronizer control unit. (L2, PDF p. 320)
- 296. b** The text compares it to control wheel steering. (L2, PDF p. 320)
- 297. c** Planned path and altitude are compared against the momentary path parameters. (L2, PDF p. 320)
- 298. a** The superposition happens inside the SCAS computer. (L2, PDF p. 320)
- 299. b** Trim signals pass through the synchronizer control unit on their way to the trim actuator. (L2, PDF p. 320)
- 300. b** Velocity vector control belongs to the cyclic stick, which works by tilting the rotor plane. (L3, PDF p. 312)
- 301. a** A sideward input leaves the front servo unenergized and moves the helicopter sideward to the left. (L3, PDF p. 314)
- 302. c** Sideward movement comes from a tilt about the longitudinal axis, produced by the LH and RH servos. (L3, PDF p. 314)
- 303. b** Changing tail rotor blade pitch changes the tail rotor thrust. (L3, PDF p. 316)
- 304. b** With speed on, a rearward cyclic input gives an airplane like climb and the speed falls off. (L3, PDF p. 319)
- 305. c** Left pedal with speed on yaws the helicopter left and costs a little speed. (L3, PDF p. 319)
- 306. b** Without inherent stability the pilot has to control the attitude all the time. (L3, PDF p. 320)
- 307. a** The main servos feed the movement back to the pilot controls through the cable system. (L3, PDF p. 320)
- 308. b** The momentary attitude becomes the new reference the stabilization system holds. (L3, PDF p. 320)
- 309. c** The rate gyro signal goes through the SCAS to the control servos, and the main servos compensate the motion. (L3, PDF p. 320)
- 310. b** The two are overlaid on each other inside the computer rather than one replacing the other. (L3, PDF p. 320)
- 311. c** The neutral stability is what the system provided, and without it the pilot is back to controlling the attitude all the time. (L3, PDF p. 320)